Thao Nguyen

Exam 3

$$70/100$$

① $A \rightarrow B$

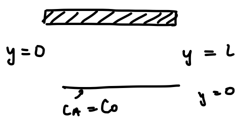
$A \rightarrow C$ Stationary liquid

$A \rightarrow F$

$C_A$ ?

$C_A|_{y=0} = C_0$

$(20/50)$




$r_{AB} = -k_B$

$r_{AC} = -k_C$

$r_{AF} = -k_F C_A$

Balance Eq.

$$\frac{d}{dy}\left(D\frac{dC_A}{dy}\right) + k_B = 0 \quad \times$$

mass balance:

$$D\frac{d^2C_A}{dy^2} - k_F C_A - k_B - k_C = 0$$

BC 1 : $C_A(y=L) = C_{AL}$

$$-D\frac{dC_A}{dy}(y=0) = -k_1 C_A|_{y=0}$$

$$C_A(y) = -\frac{k_0}{2D}y^2 + Ay + B$$

BC 1 $\Rightarrow$ $C_{AL} = -\frac{k_0}{2D}L^2 + AL + B$

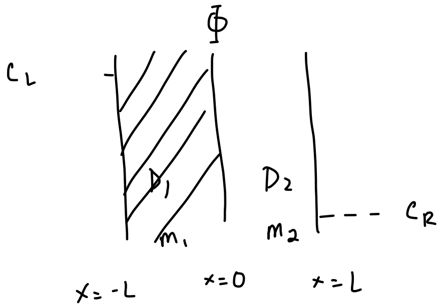
BC 2 $\Rightarrow$ $-DA = -k_1 B$ $\Rightarrow$ $B = \frac{D}{k_1}A$

$$\Rightarrow A = \frac{C_{AL} + \frac{k_0}{2D}L^2}{L + \frac{D}{k_1}} \qquad \Rightarrow B = \frac{D\left[C_{AL} + \frac{k_0}{2D}L^2\right]}{k_1 L + D} \quad \times$$

$$A = \frac{k_1\left[C_{AL} + \frac{k_0}{2D}L^2\right]}{k_1 L + D}$$

$$C_A(x) = -\frac{k_0}{2D}x^2 + \frac{k_1 x + D}{k_1 L + D}\left[C_{AL} + \frac{k_0}{2D}L^2\right] \quad \times$$

(2)




50/50

$m_1$ $\quad D_1 \dfrac{d^2 C_{A_1}}{dx^2} = 0 \quad$ for $\quad -L < x < 0$

$dC_A = C_1$

$C_{A_1} = Ax + B$

$m_2$ $\quad D_2 \dfrac{d^2 C_{A2}}{dx^2} - k_0 = 0 \quad$ for $\quad 0 < x < L$

$\dfrac{d C_{A2}}{dx} = \dfrac{k_0}{D_2} x + C_3$

$C_{A2} = \dfrac{k_0}{2D_2} x^2 + D x + E$

$BC_1$ : $C_{A_1}(x=-L) = C_L$

$BC_2$ : $C_{A_2}(x=L) = C_R$

$BC_3$ : $\phi_1 = \dfrac{C_{A_1}(x=0^-)}{C_{A2}(x=0^+)}$

$BC_4$ : $-D_1 \dfrac{dC_{A_1}}{dx}\Big|_{x=0^-} = -D_2 \dfrac{dC_{A2}}{dx}\Big|_{x=0^+}$

BC # 1 : $C_{A_1}(x=-L) = A(-L) + B = C_L \Rightarrow A = \dfrac{B-C_L}{L}$

BC # 2 : $C_{A_2}(x=L) = \dfrac{k_0}{2D_2} L^2 + DL + E = C_R \qquad E = C_R - \dfrac{k_0}{2D_2} L^2 - DL = \dfrac{B}{\phi}$

BC # 3 : $\phi_1 = \dfrac{B}{E}$

BC # 4 : $-D_1 A = -D_2 D \qquad \Rightarrow D = \dfrac{D_1 A}{D_2} = \dfrac{(B-C_L) D_1}{L D_2}$

$C_R - \dfrac{k_0}{2D_2} L^2 - \dfrac{(B-C_L)D_1}{D_2} = \dfrac{B}{\phi}$

$$C_R \phi D_L - \frac{k_0 \phi L^2}{2} - (B - C_L) D_1 \phi = B$$

$$B + (B - C_L) D_1 \phi = C_R \phi D_L - \frac{k_0 \phi L^2}{2}$$

$$B (1 + D_1 \phi) = C_R \phi D_L - \frac{k_0 \phi L^2}{2} + D_1 C_L \phi$$

$$\Rightarrow B = \frac{C_R \phi D_L - \frac{k_0 \phi L^2}{2} + D_1 C_L \phi}{1 + D_1 \phi}$$

$$\Rightarrow C_{A_1}(x) = \frac{C_R \phi D_L - \frac{k_0 \phi L^2}{2} - C_L}{(1 + D_1 \phi) L} x + \frac{C_R \phi D_L - \frac{k_0 \phi L^2}{2} + D_1 C_L \phi}{(1 + D_1 \phi)}$$

$$\Rightarrow C_{A_2}(x) = \frac{k_0}{2D_2} x^2 + \frac{C_R \phi D_L - \frac{k_0 \phi L^2}{2} - C_L}{(1 + D_1 \phi)(L)} \frac{D_1}{D_L} x + C_R - \frac{k_0}{2D_2} L^2 - DL$$

$$N''_{A_1}(x) = -D_1 \frac{dC_{A_1}}{dx} = - \frac{\left(C_R \phi D_L - \frac{k_0 \phi L^2}{2} + - C_L\right) D_1}{(1 + D_1 \phi) L}$$

$$N''_{A_2}(x) = -D_2 \frac{dC_{A_2}}{dx} = -D_2 \left( \frac{k_0}{2D_2} x + \frac{C_R \phi D_L - \frac{k_0 \phi L^2}{2} - C_L}{(1 + D_1 \phi) . L} \frac{D_1}{D_L} \right)$$